Aug 1 2025
BDBP106: Linux and Python programming
Lab 5

Learning goals: Linux OS commands grep, tar, gzip
NOTE:
For this lab, download ’BrainCancer.csv’ to a folder called ‘Lab5’. If the file download to the ”Downloads” folder by default, move the file to the folder ’Lab5’.

EXERCISES Start a fresh terminal for this Lab.
(1) Look for the pattern ‘Meningoma’ in the data file. Redirect the output to a new file called ’LinesWithMeningoma.out’. Show the command used and the contents of this new file in your screenshot.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ cd Downloads
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ cd lab5
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep Meningioma BrainCancer > LinesWithMeningoma.out
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ cat LinesWithMeningoma
cat: LinesWithMeningoma: No such file or directory
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ cat LinesWithMeningoma.out
1,Female,Meningioma,Infratentorial,90,6.11,SRS,0,57.64
3,Female,Meningioma,Infratentorial,70,7.95,SRS,0,26.46
6,Female,Meningioma,Supratentorial,80,4.82,SRS,0,52.75
7,Male,Meningioma,Supratentorial,80,3.19,SRT,0,55.8
9,Female,Meningioma,Supratentorial,70,12.16,SRT,0,34.66
12,Female,Meningioma,Infratentorial,90,6.54,SRS,0,34.26
13,Female,Meningioma,Infratentorial,90,0.63,SRS,0,32.98
15,Female,Meningioma,Supratentorial,60,9.18,SRT,0,41.44
19,Male,Meningioma,Supratentorial,80,13.49,SRS,1,6.92
20,Female,Meningioma,Supratentorial,90,2.5,SRT,0,30.16
21,Female,Meningioma,Supratentorial,80,2.82,SRS,0,24.39
24,Female,Meningioma,Infratentorial,100,2.13,SRS,1,51.02
25,Female,Meningioma,Supratentorial,70,6.48,SRT,1,33.41
29,Male,Meningioma,Supratentorial,60,3.81,SRT,0,36.1
30,Female,Meningioma,Supratentorial,90,4.72,SRS,0,65.02
32,Male,Meningioma,Supratentorial,90,2.56,SRS,0,44.39
35,Female,Meningioma,Supratentorial,90,7.3,SRT,0,5.51
37,Female,Meningioma,Supratentorial,80,6.6,SRT,0,14.75
40,Female,Meningioma,Supratentorial,90,2.54,SRT,0,45.74
41,Female,Meningioma,Supratentorial,80,1.57,SRT,0,2.03
43,Female,Meningioma,Supratentorial,70,6.7,SRT,0,14.56
44,Male,Meningioma,Supratentorial,80,12.63,SRT,1,4.16
46,Male,Meningioma,Supratentorial,60,7.09,SRS,1,31.25
48,Female,Meningioma,Supratentorial,80,26.31,SRT,1,39.54
49,Male,Meningioma,Supratentorial,70,0.97,SRT,1,1.41
52,Female,Meningioma,Infratentorial,90,9.24,SRT,0,26.85
54,Male,Meningioma,Infratentorial,80,24.41,SRT,0,39.54
60,Female,Meningioma,Supratentorial,90,2.47,SRT,0,17.57
62,Male,Meningioma,Supratentorial,80,11.51,SRT,1,14.62
64,Male,Meningioma,Supratentorial,80,4.77,SRT,0,67.38
66,Female,Meningioma,Supratentorial,100,4,SRT,0,52.23
69,Female,Meningioma,Supratentorial,80,6.93,SRS,0,10.1
70,Female,Meningioma,Supratentorial,70,3.63,SRT,0,32.82
71,Male,Meningioma,Supratentorial,70,8.45,SRT,0,19.41
72,Male,Meningioma,Supratentorial,80,20.93,SRT,1,31.15
76,Female,Meningioma,Supratentorial,80,31.74,SRT,0,57.25
77,Female,Meningioma,Supratentorial,80,2.39,SRS,0,73.74
78,Female,Meningioma,Supratentorial,90,7.26,SRT,0,49.05
79,Female,Meningioma,Supratentorial,100,9.66,SRT,0,39.25
80,Female,Meningioma,Infratentorial,70,2.94,SRS,0,1.54
86,Male,Meningioma,Supratentorial,90,2.5,SRT,0,32.82
87,Male,Meningioma,Supratentorial,90,2.02,SRS,0,42.07
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$
```

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(2) How many males and females were present in this study? How will you create a grep filter to find this out?

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -c Male BrainCancer > Male
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -c Male BrainCancer
43
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -c Female BrainCancer
45
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -c Female BrainCancer > Female
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -c -e Male -e Female BrainCancer > MaleFemale
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$
```

(3) How many males have supratentorial brain cancer and how many females have infratentorial cancer? Which option in grep makes you ignore the case while searching? Make use of this in this exercise.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -c -vi -e Female -e Infratentorial BrainCancer
35
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -c -v -e Male -e Supratentorial BrainCancer
11
```

(4) Find out which patient has the lowest ki value. (Hint: Use sort for this, it's easier)

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ sort -t ',' -k5,5 -n BrainCancer
 > lowkival
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ head -n2 lowkival
,sex,diagnosis,loc,ki,gtv,stereo,status,time
63,Female,HG glioma,Supratentorial,40,22.87,SRT,1,3.38
```

(5) Print out the output by searching for the patterns 'SRS' as well as 'infratentorial'. How many such entries are there?

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -e SRS BrainCancer > srs
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -e Infratentorial srs > srsinfra
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ cat srsinfra
1,Female,Meningioma,Infratentorial,90,6.11,SRS,0,57.64
3,Female,Meningioma,Infratentorial,70,7.95,SRS,0,26.46
12,Female,Meningioma,Infratentorial,90,6.54,SRS,0,34.26
13,Female,Meningioma,Infratentorial,90,0.63,SRS,0,32.98
23,Female,Other,Infratentorial,80,2.11,SRS,0,10.49
24,Female,Meningioma,Infratentorial,100,2.13,SRS,1,51.02
34,Male,Other,Infratentorial,80,6.81,SRS,0,57.11
56,Male,Other,Infratentorial,90,0.48,SRS,0,54.43
57,Male,HG glioma,Infratentorial,80,0.22,SRS,0,33.67
68,Male,Other,Infratentorial,70,0.01,SRS,0,23.67
80,Female,Meningioma,Infratentorial,70,2.94,SRS,0,1.54
```

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -c Infratentorial srsinfra
11
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$
```

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(6) List the patients who have survival times more than 50 months.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ sort -t ',' -k9,9 -nr BrainCancer > 50.out
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ head -n14 50.out
18,Male,HG glioma,Supratentorial,90,10.8,SRT,0,82.56
65,Male,LG glioma,Supratentorial,80,9.58,SRT,0,78.75
77,Female,Meningioma,Supratentorial,80,2.39,SRS,0,73.74
64,Male,Meningioma,Supratentorial,80,4.77,SRT,0,67.38
30,Female,Meningioma,Supratentorial,90,4.72,SRS,0,65.02
1,Female,Meningioma,Infratentorial,90,6.11,SRS,0,57.64
76,Female,Meningioma,Supratentorial,80,31.74,SRT,0,57.25
34,Male,Other,Infratentorial,80,6.81,SRS,0,57.11
7,Male,Meningioma,Supratentorial,80,3.19,SRT,0,55.8
56,Male,Other,Infratentorial,90,0.48,SRS,0,54.43
6,Female,Meningioma,Supratentorial,80,4.82,SRS,0,52.75
66,Female,Meningioma,Supratentorial,100,4,SRT,0,52.23
24,Female,Meningioma,Infratentorial,100,2.13,SRS,1,51.02
14,Male,NA,Supratentorial,90,6.38,SRT,0,50.85
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$
```

(7) List the patients who have HG glioma and have survival times more than 50 months. How many are there?

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep 'HG glioma' BrainCancer > hg_glioma
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ sort -t ',' -k9,9 -nr hg_glioma
18,Male,HG glioma,Supratentorial,90,10.8,SRT,0,82.56
```

1

(8) List the patients who have LG glioma with location supratentorial and survial times less than 30 months. How many are there?

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep 'LG glioma' BrainCancer > lg_glioma
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep Supratentorial lg_glioma > lgsupra
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ sort -t "," -k9,9 -n lgsupra > lgsuprasorted
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ head -n4 lgsuprasorted
31,Female,LG glioma,Supratentorial,80,0.85,SRS,1,6.1
50,Female,LG glioma,Supratentorial,80,0.19,SRS,0,11.51
73,Male,LG glioma,Supratentorial,90,2.64,SRT,0,20.13
26,Male,LG glioma,Supratentorial,90,4.23,SRT,1,25.02
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$
```

(9) By default does grep ignore or not ignore case? Demonstrate this by using the appropriate option and by running the grep command without this option.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -i Male BrainCancer
1,Female,Meningioma,Infratentorial,90,6.11,SRS,0,57.64
2,Male,HG glioma,Supratentorial,90,19.35,SRT,1,8.98
3,Female,Meningioma,Infratentorial,70,7.95,SRS,0,26.46
4,Female,LG glioma,Supratentorial,80,7.61,SRT,1,47.8
5,Male,HG glioma,Supratentorial,90,5.06,SRT,1,6.3
6,Female,Meningioma,Supratentorial,80,4.82,SRS,0,52.75
7,Male,Meningioma,Supratentorial,80,3.19,SRT,0,55.8
8,Male,LG glioma,Supratentorial,80,12.37,SRT,0,42.1
```

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -i male BrainCancer
1,Female,Meningioma,Infratentorial,90,6.11,SRS,0,57.64
2,Male,HG glioma,Supratentorial,90,19.35,SRT,1,8.98
3,Female,Meningioma,Infratentorial,70,7.95,SRS,0,26.46
4,Female,LG glioma,Supratentorial,80,7.61,SRT,1,47.8
5,Male,HG glioma,Supratentorial,90,5.06,SRT,1,6.3
6,Female,Meningioma,Supratentorial,80,4.82,SRS,0,52.75
7,Male,Meningioma,Supratentorial,80,3.19,SRT,0,55.8
```

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep male BrainCancer
1,Female,Meningioma,Infratentorial,90,6.11,SRS,0,57.64
3,Female,Meningioma,Infratentorial,70,7.95,SRS,0,26.46
4,Female,LG glioma,Supratentorial,80,7.61,SRT,1,47.8
6,Female,Meningioma,Supratentorial,80,4.82,SRS,0,52.75
9,Female,Meningioma,Supratentorial,70,12.16,SRT,0,34.66
12,Female,Meningioma,Infratentorial,90,6.54,SRS,0,34.26
13,Female,Meningioma,Infratentorial,90,0.63,SRS,0,32.98
15,Female,Meningioma,Supratentorial,60,9.18,SRT,0,41.44
```

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(10) List the patients who are dead at the end of the study. How many are there? Find another way to listing this set of patients (look up grep man pages to look for a specific option to select non-matching lines)
0=alive
1=dead

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -c ",1," BrainCancer
35
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -v ",0," BrainCancer
,sex,diagnosis,loc,ki,gtv,stereo,status,time
2,Male,HG glioma,Supratentorial,90,19.35,SRT,1,8.98
4,Female,LG glioma,Supratentorial,80,7.61,SRT,1,47.8
5,Male,HG glioma,Supratentorial,90,5.06,SRT,1,6.3
11,Male,LG glioma,Supratentorial,80,0.14,SRT,1,35.93
16,Female,HG glioma,Supratentorial,70,11.38,SRS,1,7.05
17,Female,Other,Infratentorial,60,24,SRT,1,6.82
19,Male,Meningioma,Supratentorial,80,13.49,SRS,1,6.92
22,Male,HG glioma,Supratentorial,70,14.44,SRT,1,14
24,Female,Meningioma,Infratentorial,100,2.13,SRS,1,51.02
25,Female,Meningioma,Supratentorial,70,6.48,SRT,1,33.41
26,Male,LG glioma,Supratentorial,90,4.23,SRT,1,25.02
27,Male,Other,Supratentorial,60,34.64,SRT,1,11.57
28,Male,HG glioma,Supratentorial,70,33.69,SRT,1,0.07
31,Female,LG glioma,Supratentorial,80,0.85,SRS,1,6.1
33,Female,Other,Infratentorial,70,13.45,SRT,1,10.82
38,Male,HG glioma,Supratentorial,90,9.95,SRT,1,6.23
39,Male,Other,Infratentorial,80,12.51,SRT,1,29.7
42,Male,HG glioma,Supratentorial,90,0.28,SRT,1,16.43
44,Male,Meningioma,Supratentorial,80,12.63,SRT,1,4.16
46,Male,Meningioma,Supratentorial,60,7.09,SRS,1,31.25
48,Female,Meningioma,Supratentorial,80,26.31,SRT,1,39.54
49,Male,Meningioma,Supratentorial,70,0.97,SRT,1,1.41
53,Male,HG glioma,Supratentorial,90,2.5,SRT,1,9.77
55,Female,HG glioma,Supratentorial,80,0.63,SRT,1,16.92
58,Male,HG glioma,Supratentorial,80,3.75,SRT,1,19.9
59,Female,Other,Supratentorial,80,11.83,SRT,1,22.03
61,Female,HG glioma,Supratentorial,80,12.08,SRT,1,7.25
62,Male,Meningioma,Supratentorial,80,11.51,SRT,1,14.62
63,Female,HG glioma,Supratentorial,40,22.87,SRT,1,3.38
67,Female,HG glioma,Supratentorial,80,7.59,SRT,1,4.56
72,Male,Meningioma,Supratentorial,80,20.93,SRT,1,31.15
74,Female,HG glioma,Supratentorial,80,0.19,SRT,1,11.02
81,Female,HG glioma,Supratentorial,80,15.45,SRT,1,46.16
84,Male,HG glioma,Supratentorial,80,0.16,SRT,1,20.69
85,Male,HG glioma,Supratentorial,80,19.81,SRT,1,6.39
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -vc ",0," BrainCancer
36
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$
```

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(11) Let's list the patients who have their cancer in infratentorial location. How many are there? How do I restrict the output to first ten patients?

```
88,Male,Other,Infratentorial,80,0.11,SRT,0,13.9
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -i Infratentorial BrainCancer
1,Female,Meningioma,Infratentorial,90,6.11,SRS,0,57.64
3,Female,Meningioma,Infratentorial,70,7.95,SRS,0,26.46
12,Female,Meningioma,Infratentorial,90,6.54,SRS,0,34.26
13,Female,Meningioma,Infratentorial,90,0.63,SRS,0,32.98
17,Female,Other,Infratentorial,60,24,SRT,1,6.82
23,Female,Other,Infratentorial,80,2.11,SRS,0,10.49
24,Female,Meningioma,Infratentorial,100,2.13,SRS,1,51.02
33,Female,Other,Infratentorial,70,13.45,SRT,1,10.82
34,Male,Other,Infratentorial,80,6.81,SRS,0,57.11
39,Male,Other,Infratentorial,80,12.51,SRT,1,29.7
45,Male,Other,Infratentorial,90,3.12,SRT,0,18.95
52,Female,Meningioma,Infratentorial,90,9.24,SRT,0,26.85
54,Male,Meningioma,Infratentorial,80,24.41,SRT,0,39.54
56,Male,Other,Infratentorial,90,0.48,SRS,0,54.43
57,Male,HG glioma,Infratentorial,80,0.22,SRS,0,33.67
68,Male,Other,Infratentorial,70,0.01,SRS,0,23.67
80,Female,Meningioma,Infratentorial,70,2.94,SRS,0,1.54
83,Male,LG glioma,Infratentorial,90,30.41,SRT,0,1.18
88,Male,Other,Infratentorial,80,0.11,SRT,0,13.9
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -ic Infratentorial BrainCancer
19
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ grep -i Infratentorial BrainCancer >
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ head -10 infracancer
1,Female,Meningioma,Infratentorial,90,6.11,SRS,0,57.64
3,Female,Meningioma,Infratentorial,70,7.95,SRS,0,26.46
12,Female,Meningioma,Infratentorial,90,6.54,SRS,0,34.26
13,Female,Meningioma,Infratentorial,90,0.63,SRS,0,32.98
17,Female,Other,Infratentorial,60,24,SRT,1,6.82
23,Female,Other,Infratentorial,80,2.11,SRS,0,10.49
24,Female,Meningioma,Infratentorial,100,2.13,SRS,1,51.02
33,Female,Other,Infratentorial,70,13.45,SRT,1,10.82
34,Male,Other,Infratentorial,80,6.81,SRS,0,57.11
39,Male,Other,Infratentorial,80,12.51,SRT,1,29.7
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$
```

(12) Create an archive file by combining the outputs of questions 1-5.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ tar -cvf q12.tar LinesWithMening
oma.out Female Male MaleSupra FemaleInfra MaleFemale lowkival srsinfra
LinesWithMeningoma.out
Female
Male
MaleSupra
FemaleInfra
MaleFemale
lowkival
srsinfra
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ ls
50.out       hg_glioma    lgsuprasorted           MaleFemale  srsinfra
BrainCancer  infracancer  LinesWithMeningoma.out  MaleSupra
Female       lg_glioma    lowkival                q12.tar
FemaleInfra  lgsupra      Male                    srs
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$
```

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(13) Create a zipped and archived file by combining the outputs of questions 6-11.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ tar -cvzf q13.tar 50.out hg_glio
ma infracancer lg_glioma lgsupra
50.out
hg_glioma
infracancer
lg_glioma
lgsupra
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ ls
50.out       hg_glioma    lgsuprasorted           MaleFemale  srs
BrainCancer  infracancer  LinesWithMeningoma.out  MaleSupra   srsinfra
Female       lg_glioma    lowkival                q12.tar
FemaleInfra  lgsupra      Male                    q13.tar
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$
```

(14) List and extract the contents of the archive file created in (12) to a new directory and list the contents of the new directory.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ mkdir q14
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ tar -xvf q12.tar -C q14
LinesWithMeningoma.out
Female
Male
MaleSupra
FemaleInfra
MaleFemale
lowkival
srsinfra
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ ls q14
Female       LinesWithMeningoma.out  Male        MaleSupra
FemaleInfra  lowkival                MaleFemale  srsinfra
```

(15) List and extract the contents of the archived-zipped file in (13) to a new directory and list the contents of the new directory.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ mkdir q15
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ tar -xvzf q13.tar -C q15
50.out
hg_glioma
infracancer
lg_glioma
lgsupra
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab5$ ls q15
50.out  hg_glioma  infracancer  lg_glioma  lgsupra
```